

SC4172 Tiny Machine Machine Learning

Tutorial 6 – System Operation Requirements

A Speech Recognition model takes 80 million MAC operations to execute one inference

- (i) What is the operating frequency of a Cortex-M4 processor required to perform an inference within 500ms?
- (ii) Given that the processor (and its peripherals) consume 3mW per inference is powered by a battery with capacity of 3000 Whr (i.e. Joules in term of energy) , how long can the system operation?
- (iii) What are the requirements of the battery to be able to support the system operation over the full duration?